

- **Push changes to the RethinkDB client:** This is the most significant concept in RethinkDB. Rather than see the changes in the database through the polling approach, RethinkDB pushes the changes in old and new values to connected clients. The clients that are connected in real-time to the database server can see the live changes in the server.
- **Query execution:** RethinkDB is perhaps the best in performing various complex computations and internal logic operations. In order to handle queries, RethinkDB divides the queries into stacks. Every stack consists of various methodologies as well as internal logic to perform operations. In order to optimise the query response, the stacks are transported to the related server and every server performs the evaluation in parallel. Then queries are combined together to get the result set, which is sent back to the client. In order to provide high performance and concurrency, RethinkDB makes use of multi-version concurrency control (MVCC). With this, every user can see a snapshot of the data, and if

any changes happen in the master copy, child copies or snapshot copies are not updated until the master copy is committed.

The sharding process in RethinkDB

RethinkDB makes use of an advanced algorithm to perform sharding operations. Sharding is done on the primary key of the table for data partitioning and doesn't make use of any other keys. In RethinkDB, the primary key and the shard key are the same.

The sharding process can take place in two ways.

- **Vertical partitioning:** Users store the data in different tables with different documents in varied databases.
- **Horizontal partitioning:** A dynamic algorithm (i.e., the Range Shard algorithm) is used to determine the break point of the table and store the data in different shards on the basis of calculations.

Features of RethinkDB

- **Changefeeds:** RethinkDB is a database designed especially for real-time applications. Using

change feeds, developers can efficiently program the database to push the latest data feeds to applications in real-time. This 'change feeds' feature overcomes the limitations of polling as the database reduces the time and complexity when responding to user queries and can support Web apps in real-time.

- **Scalability:** RethinkDB provides users with high flexibility and improved storage. The processing power of the server can be expanded in real-time by adding more servers to the cluster.
- **Powerful ReQL – a query language:** ReQL is a data-driven, abstract and advanced language to build applications. It is designed to work as a full-fledged API to change and compose all sorts of queries. All queries sent to the server are parallelised automatically on the database server and split across multiple data centres.
- **Developer friendly:** RethinkDB is very developer friendly and combines easy language with simple controls. With RethinkDB, developers can build real-time applications due to faster interactions with user responses.
- **Document-oriented structure:** It is a document database built from scratch. Working on certain objects in a database may be quite cumbersome for developers as various issues like data mapping can arise. With RethinkDB, these issues are resolved by replacing the row concept with a more flexible model called a document, as documents are objects.
- **Distributed joins:** Most of the NoSQL databases don't support joins, as the latter are not covered under the data model function but just as a function of data access. RethinkDB fully supports joins and automatically compiles them to distributed programs, executing them across clusters without any sort of manual intervention by the client.

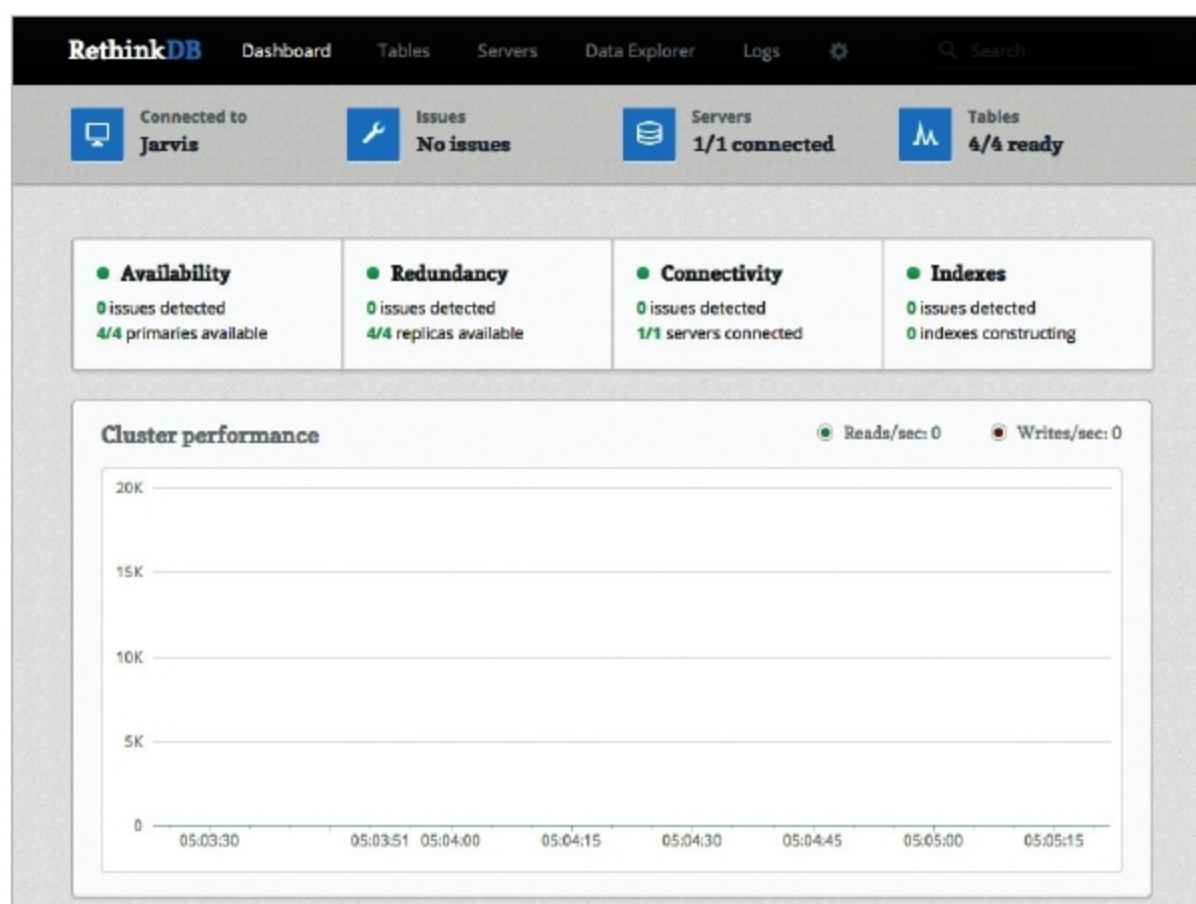


Figure 1: The RethinkDB interface